

Astable 555 Timer PCB

PCB Design Project | Tool: Eagle | 2-Layer PCB

Design Tool	Eagle
Board Type	2-Layer PCB
Power Supply	+12V in → 7805 → +5V logic

1. Project Overview

Astable 555 timer PCB with variable frequency output using 100kΩ potentiometer. NE555 IC generates continuous square wave with LED indicator. 12V input regulated via 7805. Designed in Eagle CAD.

2. Key Components

- NE555 – Timer IC (DIP-8)
- 7805 – 5V Linear Voltage Regulator (TO-92)
- Resistors: R1=10kΩ, R2=1kΩ, R3=330Ω
- R5=100kΩ Potentiometer – Frequency adjust
- Capacitors: C3/C4=10μF, C1/C2=0.1μF
- LED1 – Visual output indicator
- JP1 – 2-pin power header
- JACK – 2.0mm DC barrel jack

3. Design Features

- Astable mode: continuous square wave output
- 100kΩ potentiometer for variable frequency control
- 12V input regulated to 5V via 7805
- LED blinks at output frequency – visual feedback
- Barrel jack and pin header for power input options
- Through-hole components for easy assembly

4. Signal / Functional Flow

- [1] DC Barrel Jack / JP1 Header – 12V Input ↓
- [2] 7805 – 5V Voltage Regulator ↓
- [3] NE555 Timer IC – Astable Mode ↓
- [4] R1 (10kΩ) + R2 (1kΩ) + R5 POT (100kΩ) – RC Timing Network ↓
- [5] C3 (10μF) – Timing Capacitor ↓
- [6] Square Wave Output ↓
- [7] R3 (330Ω) + LED1 – Visual Output Indicator

5. Resume / Portfolio Description

- Designed astable 555 timer PCB in Eagle with NE555 IC, 7805 regulator, and 100kΩ potentiometer for variable frequency square wave generation.
- Board includes LED visual output indicator, barrel jack power input, and through-hole layout; Gerber files produced for PCB manufacturing.

6. Short Description (Portfolio / GitHub – ≤350 chars)

Astable 555 timer PCB with variable frequency output using 100kΩ potentiometer. NE555 IC generates continuous square wave with LED indicator. 12V input regulated via 7805. Designed in Eagle CAD.

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